Investigation of the Balmer Series

PY2107 – Experiment 7

Ian Cullinane – *125413342*

Lab Partner: Fionn Lichtsinn Doyle

Demonstrators: Jonathan and Olivia

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1 Abstract

The visible Balmer spectrum of hydrogen was measured with a diffraction grating. The diffraction angles for three spectral lines were recorded in orders one to three where visible, and the wavelengths were calculated from the grating equation. The wavelengths were then related to the initial quantum number using the Rydberg formula.

An unweighted linear fit gave an experimental hydrogen Rydberg constant of $R_H = (1.022 \pm 0.066) \times 10^7$ 1/m, where the quoted uncertainty is the standard error from the fit. This is 6.85% below the accepted value, 1.09737×10^7 1/m. The fit implies a Balmer series-limit wavelength of 391 ± 25 nm and a limit energy of 3.17 ± 0.21 eV; these are consistent with the theoretical values of approximately 364.5 nm and 3.40 eV within the fit uncertainty. The weighted fit has a reduced chi-squared of 17.91, showing that the measurement uncertainties do not account for all of the scatter. The results support the Balmer-series model, but the wavelength and Rydberg estimates remain limited by measurement scatter and systematic uncertainty.

2 Theory

The Balmer series of hydrogen arises when electrons transition from higher energy levels to the final level $n_f = 2$, emitting photons in the visible region of the electromagnetic spectrum. The wavelength of the emitted radiation is described by the Rydberg equation:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right),$$

where λ is the wavelength of the emitted photon, n_i and n_f are the initial and final principal quantum numbers, respectively, and R_H is the Rydberg constant for hydrogen.

The theoretical hydrogen Rydberg constant accounts for the finite mass of the nucleus through the reduced mass:

$$R_H = \frac{m_r}{m_e} R_\infty, \qquad m_r = \frac{m_e m_p}{m_e + m_p},$$

where m_e and m_p are the electron and proton masses, and R_∞ is the Rydberg constant for an infinitely heavy nucleus.

A diffraction grating is used to determine the wavelength of each spectral line. The grating equation is

$$d \sin\theta = \eta\lambda,$$

where d is the grating spacing, θ is the diffraction angle, and η is the diffraction order. Thus, the wavelength can be calculated using

$$\lambda = \frac{d \sin\theta}{\eta}.$$

For a grating with N lines per unit length, the spacing is $d = 1/N$.

For the linear fit, define

$$x = \frac{1}{n_f^2} - \frac{1}{n_i^2}.$$

The Rydberg equation then becomes

$$\frac{1}{\lambda} = R_H x.$$

A plot of $1/\lambda$ against x should therefore produce a straight line with gradient R_H and an ideal vertical intercept of zero. Equivalently, plotting against $1/n_i^2$ gives a gradient of $-R_H$ and an intercept of R_H/n_f^2. The analysis plot uses the first form, so its fitted gradient is the experimental Rydberg constant.

The Balmer series limit corresponds to $n_i \to \infty$. Its wavelength and transition energy are given by

$$\lambda_{\text{limit}} = \frac{n_f^2}{R_H} = \frac{4}{R_H},$$

$$E = \frac{hc}{\lambda_{\text{limit}}},$$

where h is Planck's constant and c is the speed of light.

3 Experimental Procedure

The spectrometer eyepiece was adjusted until the crosshairs were sharply focused. The telescope was then focused on a distant object until there was no parallax between the image and the crosshairs. The telescope focus was left unchanged for the remainder of the experiment.

The hydrogen discharge lamp was used to illuminate the collimator slit. The collimator lens was adjusted until the slit image was sharply focused without parallax. The slit width was reduced to produce a narrow, clearly defined image parallel to the vertical crosshair.

The diffraction grating was placed on the spectrometer table, which was levelled if necessary. The straight-through position of the telescope was recorded. The telescope was then rotated to observe spectral lines on both sides of the central image. The grating orientation was adjusted until the angular displacements on the left and right sides were approximately equal.

The angular positions of the visible Balmer spectral lines were measured on both sides of the straight-through position for as many observable diffraction orders as possible. The main and vernier scales were used to obtain angular readings. An angular reading uncertainty of 2 arcminutes was adopted for the subsequent uncertainty propagation.

The diffraction angles were determined from the measured angular positions, and the wavelength of each spectral line was calculated using the grating equation,

$$\lambda = \frac{d \sin \theta}{\eta},$$

where d is the grating spacing, θ is the diffraction angle, and η is the diffraction order.

The visible spectral lines were identified with transitions from $n_i = 3, 4$, and 5 to $n_f = 2$. No usable measurement for $n_i = 6$ was included in the analysed data.

A graph of $1/\lambda$ against $1/n_f^2 - 1/n_i^2$ was plotted. The gradient was used to determine the experimental Rydberg constant, which was compared with the theoretical value. Finally, the wavelength of the Balmer series limit and its corresponding transition energy were calculated using the experimental Rydberg constant. The hydrogen lamp was switched off after the measurements were completed. The procedure follows the laboratory instructions for this experiment.[1]

4 Analysis

The grating used had 600 lines per millimetre, giving $d = 1.667 \times 10^{-6}$ m. The 14 usable measurements comprise five observations of the $n_i = 3$ line, four of $n_i = 4$, and five of $n_i = 5$. Wavelengths were recalculated from the recorded diffraction angles and orders. For each line, the mean wavelength and its uncertainty (standard error combined in quadrature with the assumed 1% grating-spacing uncertainty) are shown in Table 1.

Table 1: Measured Balmer-line wavelengths and comparison with the theoretical values calculated using $R_H = 1.0973731568 \times 10^7$ 1/m.

$n_i \to 2$	Count	Measured λ (nm)	Theoretical λ (nm)	Difference (σ)
$3 \to 2$	5	669.79 ± 17.75	656.11	0.77
$4 \to 2$	4	515.32 ± 11.33	486.01	2.59
$5 \to 2$	5	447.56 ± 6.83	433.94	1.99

The fit variable was defined as $x = 1/n_f^2 - 1/n_i^2$, and the measured inverse wavelengths were fitted to $y = mx + c$, where $y = 1/\lambda$. The unweighted least-squares fit gave

$$m = (1.0222 \pm 0.0664) \times 10^7 \text{ 1/m}, \qquad c = (6.67 \pm 12.01) \times 10^4 \text{ 1/m}.$$

The slope is the experimental Rydberg constant. Its magnitude is 6.85% below the accepted value 1.097373×10^7 1/m. The fitted intercept is consistent with the ideal value of zero within its standard error. The fit and its extrapolation to the series limit are shown in Figure 1.

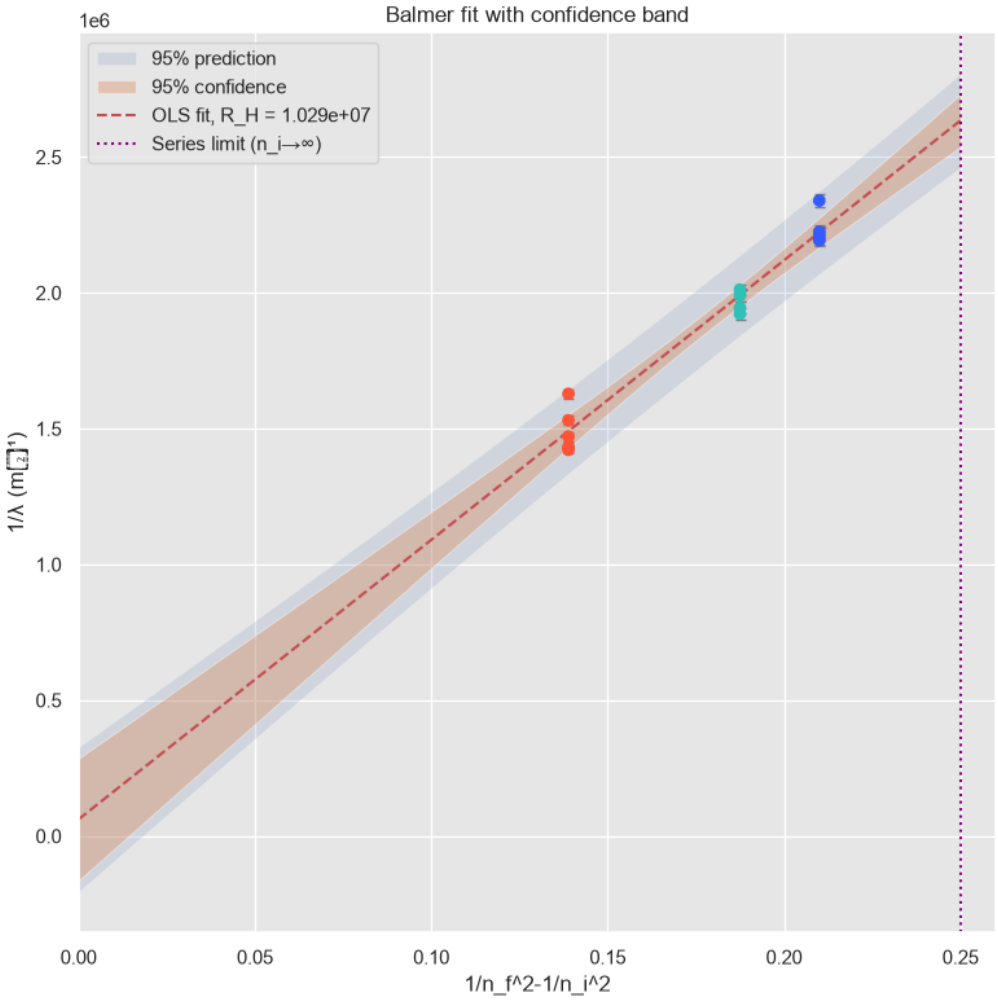


Figure 1: Inverse wavelength against $1/n_f^2 - 1/n_i^2$, with the unweighted linear fit and 95% confidence and prediction bands. The series limit occurs at $x = 1/n_f^2 = 0.25$.

For $n_i \to \infty$, $1/\lambda_{\text{limit}} = R_H/4$. Using the fitted slope gives

$$\lambda_{\text{limit}} = 391.3 \pm 25.4 \text{ nm}, \qquad E_{\text{limit}} = 3.168 \pm 0.206 \text{ eV}.$$

The theoretical values calculated from the accepted Rydberg constant are approximately 364.5 nm and 3.400 eV, respectively. These limit uncertainties are propagated from the unweighted fit and do not include the full systematic uncertainty.

The other plots provide checks on the line identification, individual wavelength measurements, and regression behaviour. The measured and theoretical wavelengths are compared in Figure 2; the measured spectrum and order-by-order wavelengths are shown in Figures 3 and 4. Figure 5 compares the unweighted, weighted, through-origin, and theoretical fits.

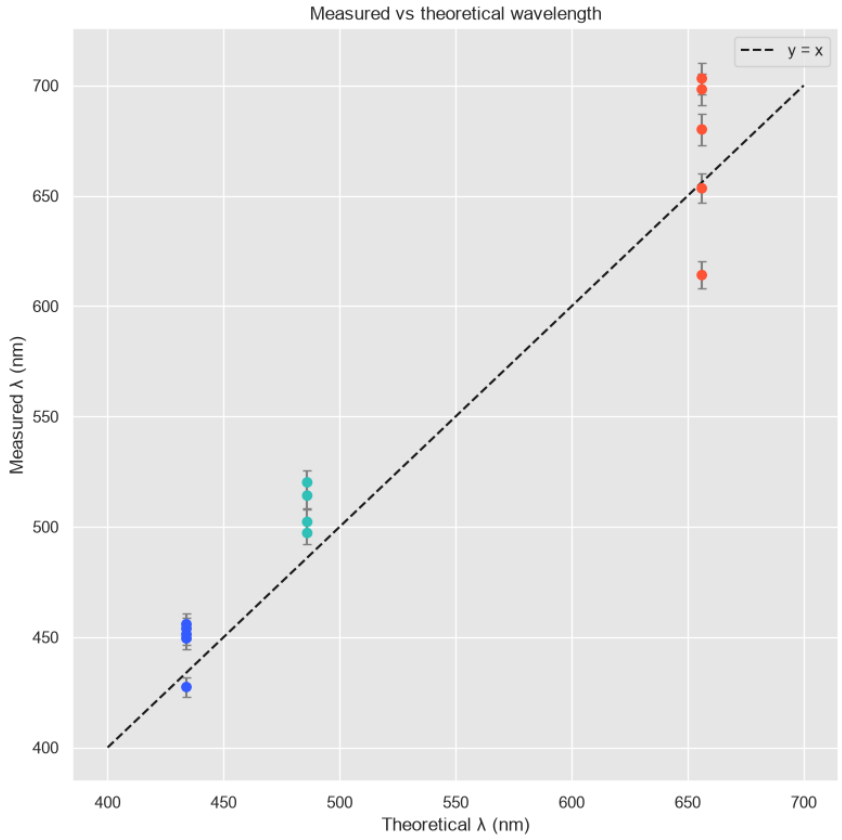


Figure 2: Measured wavelengths plotted against theoretical Balmer wavelengths. Error bars include the propagated angle uncertainty and the assumed grating-spacing uncertainty.

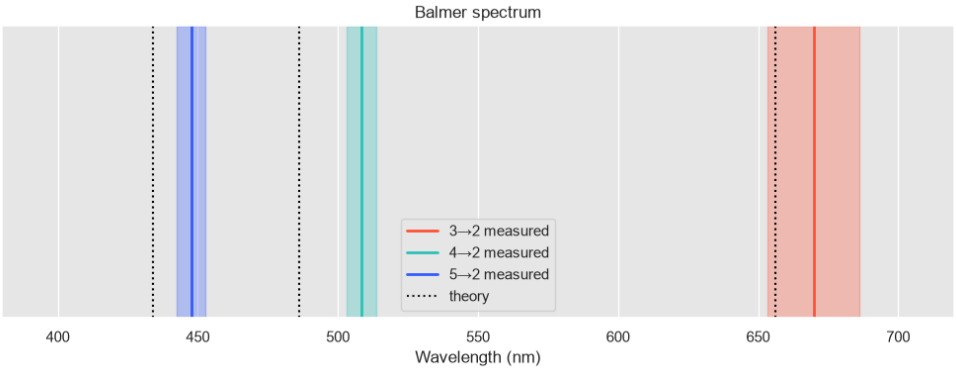


Figure 3: Measured Balmer-line positions compared with their theoretical wavelengths.

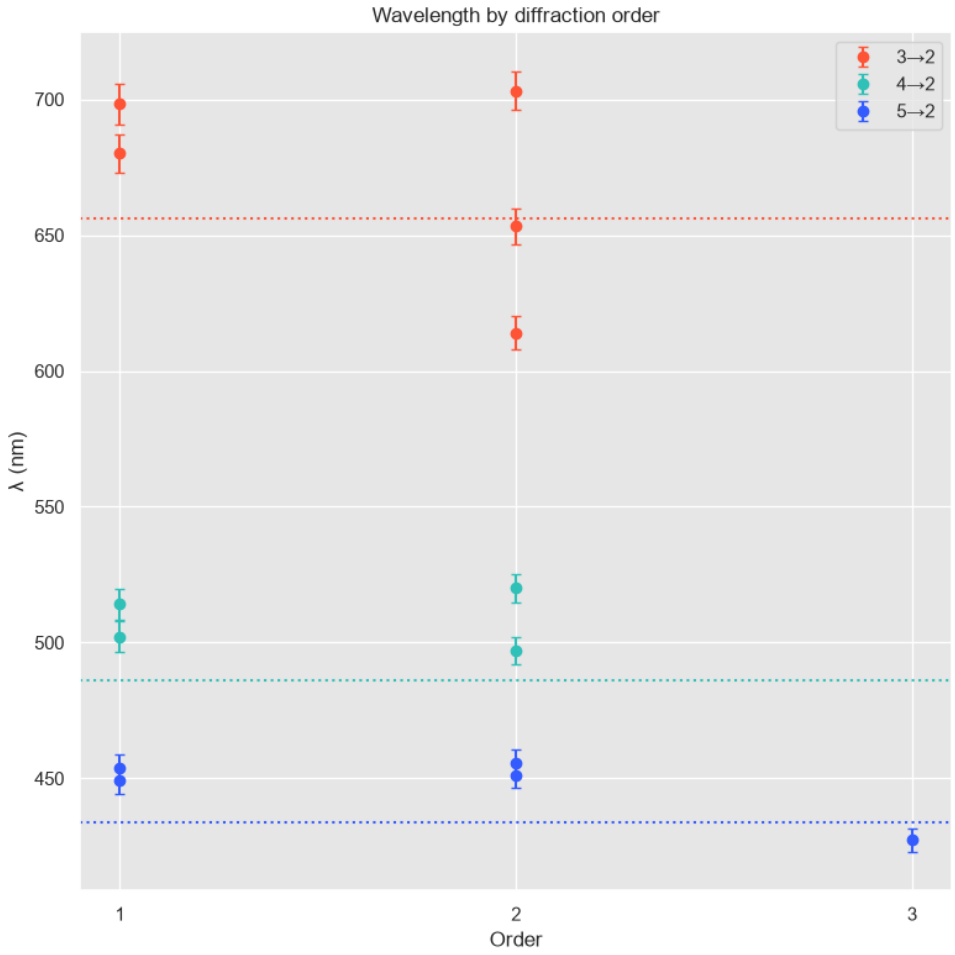


Figure 4: Wavelength estimates for each transition and diffraction order. Dotted lines indicate the theoretical wavelengths.

Table 2: Rydberg-constant estimates from alternative regressions. The weighted-fit uncertainty is formal; its goodness of fit indicates it is not an adequate description of the observed scatter.

Fit	R_H (10^7 1/m)	Reduced χ^2
Unweighted, free intercept	1.0222 ± 0.0664	–
Weighted, free intercept	1.0147 ± 0.0160	17.91
Weighted, forced through origin	1.0578 ± 0.0029	–
Accepted theoretical value	1.09737	–

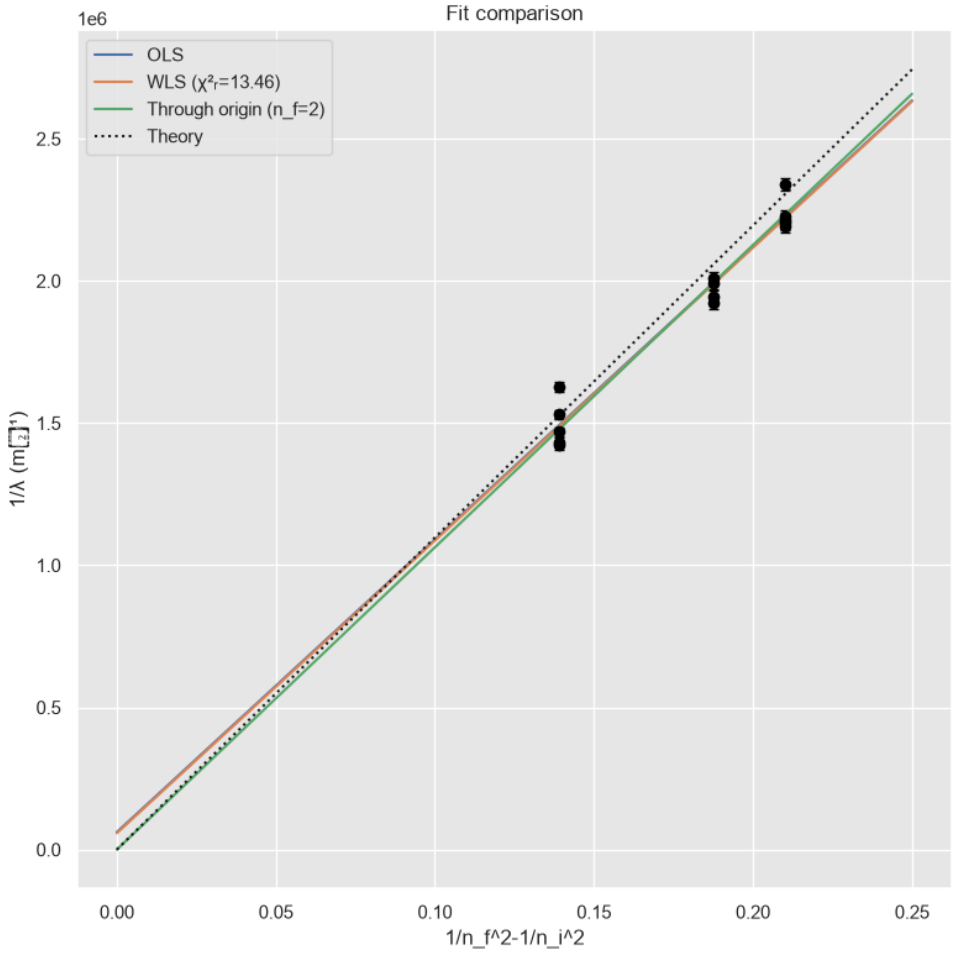


Figure 5: Comparison of regression models with the theoretical Rydberg slope. The free-intercept weighted fit has a reduced χ^2 of 17.91.

5 Error Analysis

The wavelength uncertainty was estimated by propagating an assumed angular reading uncertainty of $u_\theta = 2$ arcminutes and a 1% relative uncertainty in the grating spacing. For $\lambda = d\sin\theta/\eta$, the independent contributions were combined as

$$u_\lambda = \sqrt{\left(\frac{d\cos\theta}{\eta}u_\theta\right)^2 + (0.01\lambda)^2}, \qquad u_{1/\lambda} = \frac{u_\lambda}{\lambda^2}.$$

The fit uncertainty from the unweighted regression is 6.64×10^5 1/m, much larger than the angle-only contribution to the Rydberg uncertainty, 1.17×10^4 1/m. The assumed 1% grating spacing uncertainty contributes approximately 1.01×10^5 1/m. Thus, scatter in the observations dominates the stated fit uncertainty, although the grating calibration is a common systematic contribution that does not average away over repeated measurements.

The weighted regression gives $\chi^2 = 214.95$ for 12 degrees of freedom, corresponding to a reduced $\chi^2 = 17.91$ and a probability below 0.001 under the assumed uncertainty model. The weighted-fit error bars therefore substantially underestimate the dispersion of the measurements, or the data contain additional unmodelled effects. Consequently, the small formal weighted uncertainty should not be interpreted as the accuracy of the measurement. The residual diagnostics in Figure 6 show the deviations from the unweighted fit.

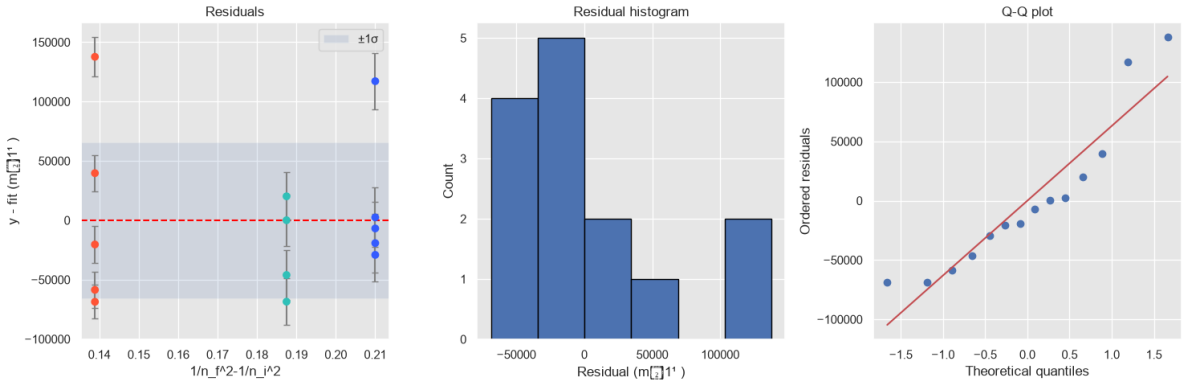


Figure 6: Unweighted-fit residuals, their distribution, and a normal-probability plot. The residual scale is larger than expected from the propagated measurement uncertainties.

6 Discussion

The observed wavelengths increase with decreasing n_i, as expected for the Balmer series, and the fitted gradient gives a Rydberg constant within approximately 1.1 unweighted-fit standard errors of the accepted value. The fitted intercept is also consistent with zero. Together, these results support the predicted linear relationship between inverse wavelength and $1/n_f^2 - 1/n_i^2$.

Each measured mean wavelength is longer than its theoretical counterpart. The $n_i = 3$ result agrees within one combined standard uncertainty, while the $n_i = 4$ result differs by about 2.6 standard uncertainties and the $n_i = 5$ result by about 2.0. The consistent direction of the offsets suggests that a common calibration or alignment effect may be present, in addition to random uncertainty. Possible contributions include uncertainty in locating the centre of a spectral line, residual error in the straight-through reference, grating alignment, and the assumed tolerance of the grating spacing.

The weighted regression's large reduced chi-squared is particularly important: the data are not described by the stated independent-error model. This explains why forcing the fit through the theoretically expected zero intercept produces an apparently very precise result despite the scatter. The unweighted fit, which reflects the observed dispersion more directly, is therefore used for the reported experimental value. The fit comparison and residual plots make this limitation visible.

The series limit is an extrapolation from the three observed lines to $n_i \to \infty$. Its uncertainty is therefore relatively large, and the reported propagated uncertainty does not fully represent the systematic and model uncertainty. The $n_i = 6$ transition was not included, so an additional violet line could not provide another independent point to constrain the fit. Future measurements could improve the result by repeating both-side angle readings, checking the angular zero and grating calibration, and including additional visible orders and spectral lines.

7 Conclusion

The Balmer spectrum was measured for transitions $n_i = 3, 4$, and 5 to $n_f = 2$. An unweighted fit to the inverse wavelengths gave $R_H = (1.022 \pm 0.066) \times 10^7$ 1/m, which is 6.85% below the accepted value. The data also gave a series-limit wavelength of 391 ± 25 nm. While the fitted slope and intercept are broadly consistent with the Rydberg model, the weighted residual test shows that the propagated measurement uncertainties do not account for all of the scatter. The result is therefore consistent with the expected Balmer-series behaviour but limited in precision by measurement scatter and systematic effects.

8 References

References

[1] Physics Department. *PY2107 : Investigation of the Balmer Series & the Rydberg constant: Experiment 7*. Laboratory handout, pp. 2.1–2.6. University Laboratory. 2026.

A Appendix

A.1 Data

Table 3: Recorded diffraction measurements and calculated wavelengths. The wavelength column is in metres.

n_i	η	θ (rad)	$\sin\theta$	λ (m)
5	1	0.273027673	0.270	4.49×10^{-7}
5	2	0.578848143	0.547	4.56×10^{-7}
5	1	0.275781414	0.272	4.54×10^{-7}
5	2	0.572293462	0.542	4.51×10^{-7}
5	3	0.877726081	0.769	4.27×10^{-7}
4	1	0.306091966	0.301	5.02×10^{-7}
4	2	0.674065550	0.624	5.20×10^{-7}
4	1	0.331050174	0.325	5.42×10^{-7}
4	2	0.639294712	0.597	4.97×10^{-7}
3	1	0.420314069	0.408	6.80×10^{-7}
3	2	0.901191063	0.784	6.53×10^{-7}
3	1	0.432279271	0.419	6.98×10^{-7}
3	2	0.828469011	0.737	6.14×10^{-7}
3	2	1.004320629	0.844	7.03×10^{-7}

A.2 Hydrogen E_n from Schrödinger Equation

$$V(r) = \frac{-e^2}{4\pi\epsilon_0 r}$$

$$\frac{-\hbar^2}{2m}\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r)$$

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}$$

$$\frac{-\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R(r)}{\partial r}\right) + \frac{\hat{L}^2}{\hbar^2 r^2}R(r)\right] + V(r)R(r) = ER(r)$$

$$\frac{\partial\Psi}{\partial r} = Y\frac{\partial R}{\partial r}$$

$$\hat{L}^2\Psi = R\hat{L}^2Y$$

$$-\frac{\hbar^2}{2m}\frac{1}{R(r)r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R(r)}{\partial r}\right) + \frac{\hat{L}^2}{2mr^2} = E$$